BUS43-2

Operations & Supply Chain Management

Diagnosing, piloting, and running operations at scale

Would the strategy, marketing, and finance plan hold once this system has to deliver it?

Albert School — 22 hours

Contents

1 Supply chain structure and coordination

A supply chain is the multi-tier network of suppliers, producers, distributors, and customers through which material and information flow. Building on the stock-and-flow view from BUS13-2, each tier holds inventory (a stock) and is connected to adjacent tiers by flows of goods and orders.

1.1 Tiers and lead times

Number the tiers from the end customer upstream: tier 0 is the retailer, tier 1 its supplier, and so on. The **lead time** of a tier is the elapsed time between placing an order on its upstream supplier and receiving the goods. Total replenishment lead time for the chain is the sum of the per-tier lead times plus any processing delays.

1.2 Information flows

Two flows run in opposite directions: goods move downstream (toward the customer), orders and demand information move upstream. A tier that sees only the orders of its immediate downstream customer — not true end-customer demand — has **local information only**.

1.3 The bullwhip effect

> **Definition 1** The **bullwhip effect** is the amplification of demand variability as orders move upstream: the variance of orders placed by a tier is larger than the variance of demand it observes, and this amplification compounds tier by tier.

Principal causes are demand-signal processing (re-forecasting from orders rather than end demand), order batching, price fluctuation and promotion-driven forward-buying, and rationing/shortage gaming.

> **Proposition 2** Structural interventions that dampen amplification include sharing end-customer demand data across tiers, reducing batch sizes and lead times, stabilising prices, and allocating scarce supply on a basis that removes the incentive to inflate orders.

2 Platform and digital operations

The course treats physical supply chains and digital platforms together because the systemic principles are identical: both move flows through capacity-limited stages and both fail when demand exceeds what a stage can serve. The operational backbone of a digital business is built from the following components.

- **API rate limits.** A cap on the number of requests a client may make per unit time. Rate limits protect a service from overload; they are the digital analogue of a capacity constraint and must be provisioned against expected and peak call volume.
- **Data pipeline reliability.** The probability that a pipeline ingesting, transforming, and delivering data completes correctly and on time. Pipelines are multi-stage flows; a failure or delay at one stage propagates downstream.
- **Marketplace liquidity.** The degree to which buyers reliably find sellers (and vice versa) quickly. A two-sided marketplace must match supply and demand on both sides; thin liquidity on either side degrades the service.
- **Content moderation at scale.** The operation of reviewing user-generated content for policy violations as volume grows. It mixes automated classification with human review and is bounded by both throughput and error-cost.

3 Demand-supply matching

The operational problem is to match uncertain demand against provisioned supply at an acceptable cost and service level.

3.1 Forecasting fundamentals

A forecast is an estimate of future demand. Every forecast carries error; the relevant quantity for operations is not the point forecast but the **distribution** of demand, summarised by its mean and its standard deviation σ over the relevant horizon.

3.2 Safety stock (physical)

Definition 3 **Safety stock** is inventory held above expected demand to absorb demand and lead-time variability and so meet a target service level.

For a target cycle service level with z-value z, demand standard deviation σ_d per period, and lead time L periods (demand variability only):

$$\text{Safety stock} = z \cdot \sigma_d \cdot \sqrt{L}.$$

The reorder point is then $\text{ROP} = d \cdot L + z\sigma_d\sqrt{L}$, where d is mean demand per period.

3.3 Capacity provisioning (digital)

The digital counterpart of safety stock is **capacity provisioning**: holding serving capacity (servers, request budget) above expected load to absorb traffic variability and meet a target availability. Provisioned headroom plays the role of safety stock; the utilization-delay relationship from BUS13-2 governs how much headroom a given latency target requires.

3.4 Service level trade-offs

Definition 4 The **service level** is the probability that demand is met from available stock or capacity within the promised time.

Raising the service level requires more safety stock or capacity, and cost rises faster than service level near the top of the range because z grows steeply as the target approaches 100%. The matching decision sets service level where the marginal holding/provisioning cost equals the marginal cost of a stockout or outage.

4 Reliability vs. efficiency

Efficiency pushes utilization up and slack out; reliability requires slack and redundancy. The two trade off, and a system cannot be maximally efficient and maximally reliable at once.

4.1 Classifying components

Classify every component of an operational system by the consequence of its failure:

- **must never fail** — failure is catastrophic or unrecoverable; justify with the failure cost;
- **can degrade gracefully** — failure is tolerable if the system continues at reduced service;
- **can fail cheaply** — failure is cheap to absorb or recover from.

The classification is justified by **failure-cost estimates**, not intuition.

4.2 Normal Accident Theory

Definition 5 **Normal Accident Theory** (Perrow) holds that in systems with high **interactive complexity** (unexpected interactions between components) and **tight coupling** (little slack or buffering between steps), serious accidents are inevitable — "normal" — because no operator can foresee every interaction and tight coupling gives no time to intervene.

The two dimensions, coupling and complexity, form the framework used to analyse operational incidents (Chapter coverage of outage, disruption, and quality failures).

4.3 High-Reliability Organisations

Definition 6 **High-Reliability Organisations** (HROs) are organisations that sustain very low failure rates in hazardous settings through preoccupation with failure, reluctance to simplify interpretations, sensitivity to operations, commitment to resilience, and deference to expertise.

4.4 Graceful degradation design

Definition 7 **Graceful degradation** is designing a system so that when a component fails the system continues at reduced capability rather than failing completely.

Applied to physical systems (factory lines that reroute around a stopped station, logistics networks that reassign load) and digital systems (payment processing that queues and retries, a platform that sheds non-essential features to preserve uptime).

5 Scaling failure modes

The question is what breaks when current volume is multiplied by $2\times$, $5\times$, and $10\times$.

5.1 Sublinear vs. superlinear cost growth

For a volume multiplier k, a cost that grows as k^α is **sublinear** when $\alpha < 1$ (economies of scale), **linear** when $\alpha = 1$, and **superlinear** when $\alpha > 1$ (cost grows faster than volume). Coordination and congestion costs are typically superlinear and are the usual source of scaling failure.

5.2 Phase transitions

Definition 8 A **phase transition** is a volume threshold at which system behaviour changes qualitatively rather than gradually — for example, a queue that is stable below a utilization threshold and unbounded above it.

5.3 The success disaster

Definition 9 A **success disaster** is a failure caused by demand exceeding what the operation was built to serve — the system breaks precisely because the strategy, marketing, or sales succeeded.

A scaling analysis identifies the first failure points, the causal chain at each, and whether each failure scales linearly or superlinearly.

6 Coordination costs

Coordination is the work of keeping interdependent parts of an operation aligned. It is itself an operational variable with measurable cost.

6.1 Handoff analysis

A **handoff** is a transfer of work or responsibility between people or teams. Each handoff adds delay, queueing, and the risk of dropped context. The **coordination tax** is the total person-hours consumed by handoffs, approvals, and dependencies in a process.

6.2 Decision latency

Definition 10 **Decision latency** is the elapsed time between a decision becoming necessary and it being made and acted on. Approvals and multi-party handoffs increase it.

6.3 Brooks's Law

Definition 11 **Brooks's Law**: adding people to a late project makes it later, because new members consume ramp-up time and the number of communication channels among n people grows as $\frac{n(n-1)}{2}$.

6.4 Conway's Law

Definition 12 **Conway's Law**: a system's structure mirrors the communication structure of the organisation that builds it. Consequently, **team topology** — how teams are drawn and connected — is an operational design variable, because it determines the handoffs and interfaces the system will have.

7 Automation boundaries

7.1 The automation spectrum

Automation is a spectrum, not a binary:

1. **manual** — the human performs the task;
2. **decision support** — the system advises, the human decides and acts;
3. **human-in-the-loop** — the system acts only with human approval of each action;
4. **human-on-the-loop** — the system acts autonomously, the human monitors and can intervene;
5. **full automation** — the system acts without human involvement.

7.2 Bainbridge's ironies of automation

Definition 13 **Bainbridge's ironies of automation**: automating the routine leaves humans the hardest cases with the least practice, and the better the automation, the more the operator's skills atrophy — so the human is least able to take over exactly when needed.

7.3 Escalation design and accountability

Every automation above the manual level needs an **escalation protocol**: the explicit rule for when control passes to a human, who that human is, and what context they receive. Under algorithmic decision-making, **accountability** must be assigned explicitly, because responsibility for an automated decision does not attach to anyone by default. The RACI matrix (Chapter on the feasibility audit uses the same tool) makes this assignment for a mixed human–AI process and exposes where accountability gaps emerge if the AI fails silently.

8 Operational debt

Definition 14 **Operational debt** is accumulated shortcut and neglect in how an operation runs, which raises its future cost — the operational analogue of technical debt.

Three kinds:

- **workaround debt** — recurring manual fixes that substitute for a real solution;
- **process debt** — processes that no longer fit the work but persist;
- **knowledge debt** — undocumented, person-dependent knowledge.

Detection signals include rising rates of manual intervention, recurring incidents with the same root cause, onboarding that depends on specific individuals, and steps no one can explain. The **refactoring decision** weighs the compound cost of leaving the debt against the cost of fixing it, and prioritises the debts whose compounding cost is highest.

9 Cross-pillar feasibility audit

The course's integrating task is the **operational feasibility audit**: taking a combined strategy, marketing, and finance plan and evaluating whether the operation can deliver it.

Procedure:

1. Extract every operational assumption embedded in the plan (volumes, lead times, service levels, growth rates, automation claims).
2. Classify each assumption as **realistic**, **optimistic-but-achievable**, or **structurally impossible**.
3. For each non-realistic assumption, name what would break, using the scaling, coordination, reliability, and debt analyses from the preceding chapters.
4. Report the operational feasibility verdict to the strategy, marketing, and finance counterparts in their terms.

This is the operational feasibility layer of the program: every plan from the other pillars is checked against whether the system that must deliver it can actually do so at the required scale, reliability, and cost.

10 Exercises

Exercises

1. Map a multi-tier supply chain and identify the tier at which demand variability is most amplified. Propose two structural interventions that dampen the bullwhip effect and justify each.

2. For a product with mean daily demand $d = 200$, demand standard deviation $\sigma_d = 40$, lead time $L = 9$ days, and a target service level with $z = 1.65$, compute the safety stock and the reorder point. Explain the cost–service-level trade-off to a finance counterpart.

3. Classify every component of a named operational system as "must never fail," "can degrade gracefully," or "can fail cheaply," justifying each with a failure-cost estimate.

4. Analyse a real operational incident using Perrow's coupling–complexity framework. Identify the design decisions that made the failure possible.

5. Stress-test a business operation at 2×, 5×, and 10× current volume. Identify the first three failure points, the causal chain at each, and whether each failure scales linearly or superlinearly.
6. Map every handoff, approval, and dependency in a real business process, calculate the coordination tax in person-hours, and redesign to reduce handoffs by at least 30% without losing accountability.
7. Place an automation proposal on the spectrum from manual to full automation and design the escalation protocol for when the automation fails.
8. Conduct an operational-debt audit of a real or simulated system: identify workaround, process, and knowledge debt, estimate the compound cost, and propose a prioritised refactoring plan.
9. Build a RACI matrix for a mixed human–AI operational process and identify where accountability gaps emerge if the AI component fails silently for 24 hours.
10. Evaluate a combined strategy + marketing + finance plan for operational feasibility. Classify each embedded assumption as realistic, optimistic-but-achievable, or structurally impossible, and name what would break.